

AssignmentNo:12

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Roll No:14 -C

Java Code

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class LogAnalysis {
    public static class LogMapper
        extends Mapper<Object, Text, Text, IntWritable>
    {
        private final static IntWritable one = new IntWritable(1);
        private Text logLevel = new Text();
        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {
            String line = value.toString();
            String[] words = line.split(" ");
            if (words.length > 2) {
                String level = words[2]; // INFO, ERROR, WARNING
                logLevel.set(level);
                context.write(logLevel, one);
            }
        }
    }

    public static class LogReducer
        extends Reducer<Text, IntWritable, Text, IntWritable> {
        private IntWritable result = new IntWritable();
        public void reduce(Text key, Iterable<IntWritable> values,
            Context context)
            throws IOException, InterruptedException {
            int sum = 0;
            for (IntWritable val : values) {
                sum += val.get();
            }
            result.set(sum);
            context.write(key, result);
        }
    }
}
```

}
}
;

```

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Log Analysis");
    job.setJarByClass(LogAnalysis.class);
    job.setMapperClass(LogMapper.class);
    job.setReducerClass(LogReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

Output

```

student@hadoop:~/loganalysis$ hadoop jar loganalysis.jar LogAnalysis logs.txt output ← Run MapReduce Log Analysis Job
2025-05-26 12:10:15,123 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
2025-05-26 12:10:15,456 INFO input.FileInputFormat: Total input files to process : 1
2025-05-26 12:10:15,789 INFO mapreduce.JobSubmitter: number of splits:1
2025-05-26 12:10:16,001 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1748242405123_0001
2025-05-26 12:10:16,412 INFO impl.YarnClientImpl: Submitted application application_1748242405123_0001
2025-05-26 12:10:16,834 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1748242405123_0001/
2025-05-26 12:10:16,835 INFO mapreduce.Job: Running job: job_1748242405123_0001
2025-05-26 12:10:22,231 INFO mapreduce.Job: Job job_1748242405123_0001 running in uber mode : false
2025-05-26 12:10:22,232 INFO mapreduce.Job: map 100% reduce 100%
2025-05-26 12:10:23,146 INFO mapreduce.Job: Job job_1748242405123_0001 completed successfully
2025-05-26 12:10:23,146 INFO mapreduce.Job: Counters: 49
  File System Counters
    FILE: Number of bytes read=195
    FILE: Number of bytes written=671246
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=310
    HDFS: Number of bytes written=133
    HDFS: Number of read operations=6
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
  Job Counters
    Launched reduce tasks=1
student@hadoop:~/loganalysis$ cat output/part-r-00000 ← View Output
ERROR      2
INFO       2
WARNING    1
student@hadoop:~/loganalysis$

```

← Log Level Count Output